Marmara Üniversitesi
Electrical and Electronics Engineering Department
EE4051: Electromagnetic Waves (Spring 2024)

Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Phone	Office Hours MTWT: 13:00-14:00	Email: mohammad.alsunaidi@marmara	
Course Information	Text: M. Sadiku *Elements of Electromagnetics*	Grading	Class Work 30%		Midterm Exam 30%	Final Exam 40%

Week	Events	Subjects	Wednesdays	Fridays
1 Feb 12-16	---	Electromagnetic Waves: Theory and Analysis	Introduction to Electromagnetics.	Waves in General. Examples of EM Waves in One dimension
2 Feb 19-23	---		The Wave Equation and Solution, Types of Waves.	Representation of EM waves in 1D. Wave properties and Parameters.
3 Feb 26-Mar 1	Course drop/add Week		EM Spectrum, Wave-matter Interactions, Classification of Materials.	Maxwell's equations. Classification of EM Wave Problems.
4 Mar 4-8	---	Reflection, Refraction, Transmission and Absorption	Normal and Oblique Propagation.	TE and TM Representations.
5 Mar 11-15	Quiz 1 March 15 (In-class)		Single Interface at Normal Incidence.	Fresnel Equations. Examples.
6 Mar 18-22	---		Single Interface at Oblique Incidence.	Multiple Interfaces.
7 Mar 25-29	---		Interference and Anti Reflection Coating. Scattering	Wave-matter interaction. Absorption.
April 1-7	**Mid-term Exams period**			
Apr 9-12 Holidays				
9 Apr 15-19	---	EM Wave Guiding	Transmission Lines	Transmission Lines
10 Apr 22-26	Midterm Make-up (Excused, Week 11)		Transmission Lines	Transmission Lines
11 Apr 29-May 3	Labor Day: May 1		Waveguides	Waveguides
12 May 6-10	---		Cavities	Optical fibers
13 May 13-17	Quiz 2 May 17 (In-class)	EM Wave Generation	The Basic Hertzian Dipole Antenna.	Monopole and Half-Wave Dipole Antennas.
14 May 20-24	YS Day May 19		The Patch Antenna	Other Antenna Types
15 May 27-31	---	Special Topics	Topic 1	Topic 2
June 3-14	**Final Exams period**		**July 1-7 (Re-sit Final Exams period)**	